

AP Calculus II
Notes 10.2
Parametric Equations

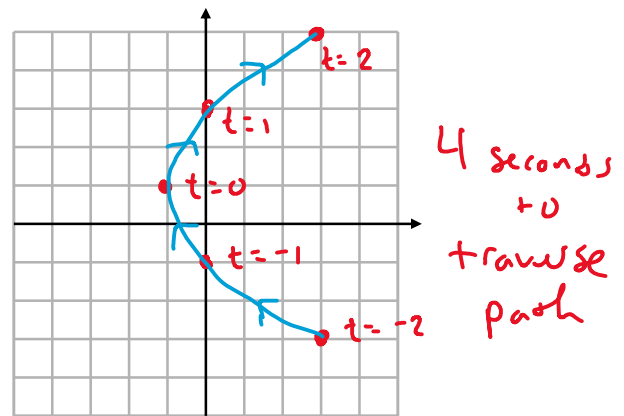
Definition of a Plane Curve: If f and g are continuous functions of t on an interval I , then the equations

$$x = f(t) \quad \text{and} \quad y = g(t)$$

are called **parametric equations** and t is called a **parameter**, usually measuring time. The set of ordered pairs (x, y) obtained as t varies over the interval I is called the **graph** of the parametric equations. Taken together, the parametric equations and the graph are called a **plane curve**, denoted by C . This essentially allows us to determine where the x and y coordinates are over time through the new variable t .

Ex. 1: Sketch the curve denoted by the parametric equations $x = t^2 - 1$ & $y = 2t + 1$.

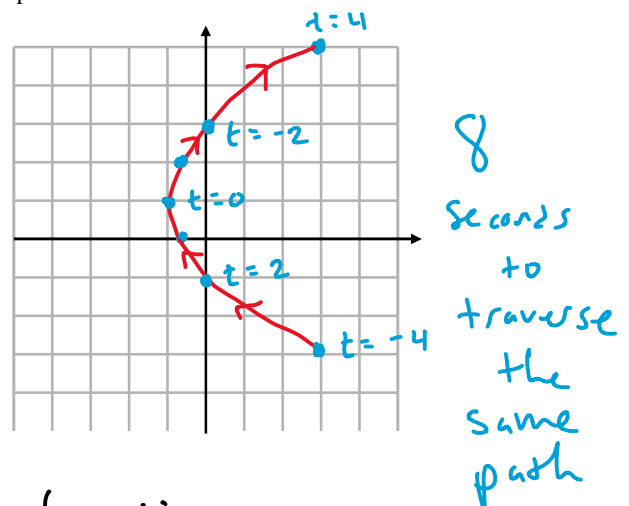
t	x	y
-2	3	-3
-1	0	-1
0	-1	1
1	0	3
2	3	5



The importance of parametric equations is usually found in time and speeds. Here is a good example of how speeds can be determined/manipulated.

Ex. 2: Sketch the curve denoted by the parametric equations $x = \frac{1}{4}t^2 - 1$ & $y = t + 1$.

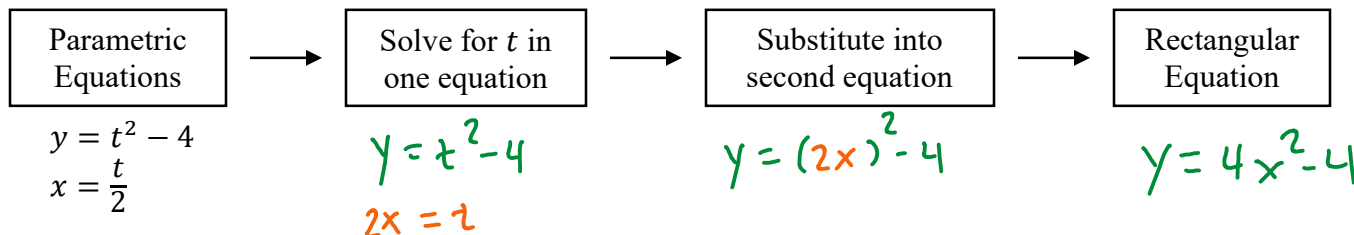
t	x	y
-4	3	-3
-2	0	-1
-1	-7/5	0
0	-1	1
1	-7/5	2
2	0	3
4	3	5



we were able to increase the time it takes to go over the same path

Eliminating the Parameter

Finding a rectangular equation that represents the graph of a set of parametric equations is called **eliminating the parameter**. Sometimes it is desirable to eliminate the parameter to recognize the curve.



Ex. 3: Rewrite the following equation in parametric form into the rectangular form.

could be $y = f(x)$ or $x = f(y)$

$x = \frac{1}{\sqrt{t+1}}$ and $y = \frac{t}{t+1}, t > -1$

easier to solve this $\rightarrow$

$\sqrt{t+1} = \frac{1}{x}$
 $t+1 = \frac{1}{x^2}$
 $t = \frac{1}{x^2} - 1$

$y = \frac{\frac{1}{x^2} - 1}{\frac{1}{x^2} - 1 + 1} = \frac{\frac{1}{x^2} - 1}{\frac{1}{x^2}} \cdot \frac{x^2}{x^2}$

$y = \frac{1 - x^2}{1} = \boxed{1 - x^2}$

Ex. 4: Rewrite the following equation in parametric form into the rectangular form and then sketch.

$$x = 5 \sin \theta \text{ and } y = 5 \cos \theta, 0 \leq \theta \leq 2\pi$$

we could solve for θ ; $\frac{x}{5} = \sin \theta, \theta = \arcsin(\frac{x}{5})$

$y = 5 \cos(\arcsin(\frac{x}{5}))$ ← NOT A NICE FUNCTION

Instead, let's try to eliminate θ another way...

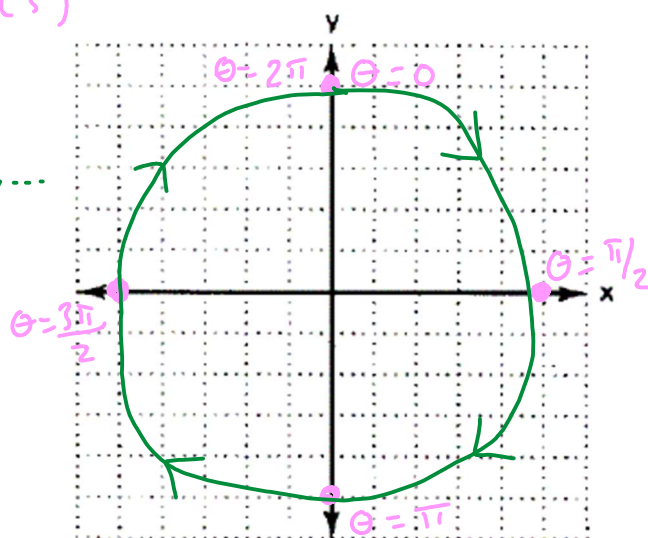
$(x = 5 \sin \theta)^2, (y = 5 \cos \theta)^2$

$\frac{x^2}{25} = \sin^2 \theta, \frac{y^2}{25} = \cos^2 \theta$

$\ast \sin^2 \theta + \cos^2 \theta = 1 \ast$

$\frac{x^2}{25} + \frac{y^2}{25} = 1 \rightarrow$

$\boxed{x^2 + y^2 = 25}$ ✓
CIRCLE!



θ	0	$\pi/2$	π	$3\pi/2$	2π
x	0	5	0	-5	0
y	5	0	-5	0	5

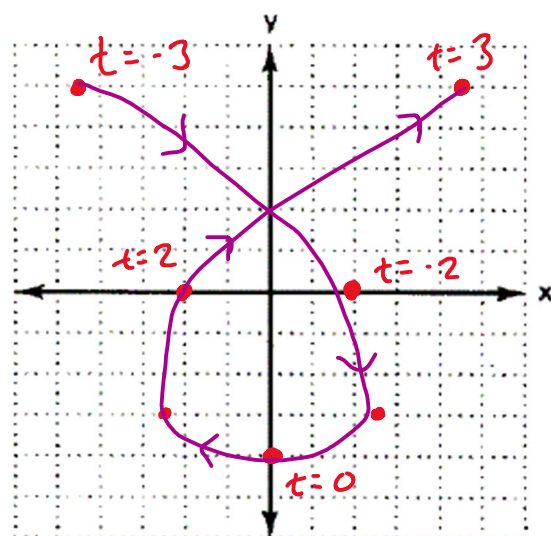
Ex. 5: Sketch the following parametric equation
(including orientation and labeling)
from $-3 \leq t \leq 3$:

$$x = \frac{1}{2}t^3 - 3t$$

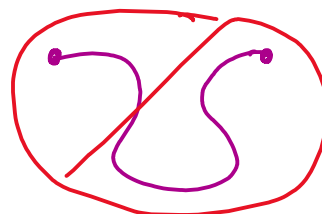
$$y = t^2 - 4$$

t	x	y
-3	-9/2	5
-2	2	0
-1	5/2	-3
0	0	-4
1	-5/2	-3
2	-2	0
3	9/2	5

Look at
how
(x,y)
changes
as
t
increases!



NOT a game of "connect the dots"

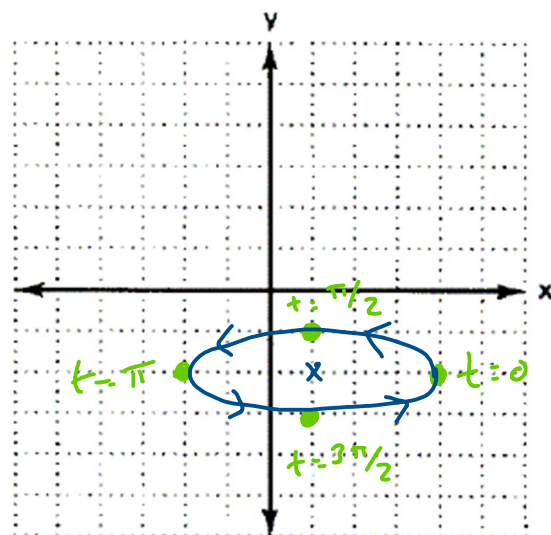


Ex. 6: Sketch the following parametric equation
(including orientation and labeling)

$$x = 1 + 3 \cos \theta$$

$$y = -2 + \sin \theta$$

θ	x	y
0	4	-2
$\pi/2$	1	-1
π	-2	-2
$3\pi/2$	1	-3
2π	4	-2



$$\frac{(x-1)}{3} = \cos \theta, \quad \frac{y+2}{1} = \sin \theta$$

$$\frac{(x-1)^2}{9} = \cos^2 \theta, \quad \frac{(y+2)^2}{1} = \sin^2 \theta$$

$$\frac{(x-1)^2}{9} + \frac{(y+2)^2}{1} = 1 \quad c: (1, -2)$$

Ellipse:

Ex. 7: The left field fence at Camden Yards is 337 feet from home plate and is 7 feet high. Chris Davis hits the baseball at a point 3 feet about the ground. It leaves the bat at an angle of 41° and at a speed of 107 feet per second. h

$\checkmark$ Given $x = (v_0 \cos \theta)t$ $y = h + (v_0 \sin \theta)t - 16t^2$

a) Write a set of parametric equations for the path of the baseball.

$$x = (107 \cos 41^\circ)t$$

$$y = 3 + (107 \sin 41^\circ)t - 16t^2$$

b) How high is the baseball after 2 seconds in flight?

that's a y-value!

$$y(2) = 79.397 \text{ ft}$$

c) How long will it take for the ball to reach its maximum height?

max y-value!

$$y' = 107 \sin 41^\circ - 32t = 0$$

$$t = \frac{107 \sin 41^\circ}{32} = 2.194 \text{ s}$$

d) Did Chris Davis hit a homerun? Be sure to explain/justify your answer.

$$x = 337 \text{ ft} \quad 337 = (107 \cos 41^\circ)t \quad t = 4.173 \text{ sec}$$

$$y(4.173171764) = 17.303 \text{ ft high}$$

$$17.303 > 7$$

$\therefore \text{Yes!}$

AP Calculus II
Notes 10.3
Parametric Equations and Calculus

The slope of the line tangent to a curve described by $x = f(t)$ and $y = g(t)$ is the first derivative of y with respect to x . However, both x and y are in terms of t , so the slope of the tangent line is defined as

$$\boxed{\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}} = \frac{dy/dt \cdot \frac{dt}{dx}}{\frac{dx}{dt}} = \frac{dy}{dx} \checkmark \quad \text{+ not a quotient rule +}$$

The parametric curve will have a horizontal tangent when $\frac{dy}{dt} = 0 \quad \left(\frac{dx}{dt} \neq 0\right)$

↳ makes sense because $\frac{dy}{dt} = 0$ means y 's aren't changing as x 's change

The parametric curve will have a vertical tangent when $\frac{dx}{dt} = 0 \quad \left(\frac{dy}{dt} \neq 0\right)$

↳ makes sense because $\frac{dx}{dt} = 0$ means x 's aren't changing as y 's change

Ex. 1: For the curve $x(t) = t^3$, $y(t) = t^2 - 2t$

a) Find the equation of the tangent line at the ordered pair $(8,0)$.

we need point + slope

Slope: $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{y'(t)}{x'(t)} = \frac{2t-2}{3t^2} \leftarrow \text{in terms of } t \text{ not } x \text{ + } y$

$x=8=t^3, t=2$ and $0=y=t^2-2t, 0=t(t-2) \quad t=0, 2$

$\therefore t=2 \quad \frac{dy}{dx} \Big|_{t=2} = \frac{2}{12} = \frac{1}{6} \rightarrow \boxed{y-0 = \frac{1}{6}(x-8)}$

b) Find the points of the horizontal and vertical tangents to the curve.

$\frac{dy}{dt} = 0 \quad \rightarrow \quad \frac{dx}{dt} = 0$

$y' = 2t - 2 = 0$
 $t = 1$

$x' = 3t^2 = 0$
 $t = 0$

$(x(1), y(1)) = \boxed{(1, -1)}$

$(x(0), y(0)) = \boxed{(0, 0)}$

Second Derivative and Concavity

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left[\frac{dy}{dx} \right] = \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{\frac{dx}{dt}}$$

since $\frac{dy}{dx}$ is in terms of t ,
 we differentiate with respect to t
 ← then divide by $\frac{dx}{dt}$

Ex. 2: Consider the parametric equation $x(t) = 3 + \sin \pi t$, $y(t) = 1 + e^{2t}$.

a) Find $\frac{dx}{dt}$, $\frac{dy}{dt}$ and $\frac{dy}{dx}$ at $t = 1.6$. Then, explain the graphical meaning of each.

$$\frac{dx}{dt} = \pi \cos \pi t, \quad x'(1.6) = 0.971 \rightarrow \text{Since } x'(1.6) > 0, \text{ the graph moves to the right}$$

$$\frac{dy}{dt} = 2e^{2t}, \quad y'(1.6) = 49.065 \rightarrow \text{Since } y'(1.6) > 0, \text{ the graph is moving up}$$

$$\frac{dy}{dx} = \frac{y'}{x'} = \frac{2e^{2t}}{\pi \cos \pi t}, \quad \left. \frac{dy}{dx} \right|_{t=1.6} = 50.541 \rightarrow \text{Since } \frac{dy}{dx} > 0, \text{ the graph is increasing}$$

b) Determine if the equation is increasing or decreasing and concave up or down at $t = 1.6$.

Since $\frac{dy}{dx} > 0$, the graph is increasing

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{\frac{dx}{dt}} \leftarrow \text{quotient rule!!}$$

$$\frac{\left[\frac{(\pi \cos \pi t)(4e^{2t}) - (2e^{2t})(-\pi^2 \sin \pi t)}{(\pi \cos \pi t)^2} \right]}{\pi \cos \pi t}$$

$$\frac{d^2y}{dx^2} = \frac{4\pi e^{2t} \cos \pi t + 2\pi^2 e^{2t} \sin \pi t}{(\pi \cos \pi t)^3} \bigg|_{t=1.6} = \frac{-365.286}{.914948} = -399.24320$$

So concave down

Ex. 3: Find the first two derivatives of the function determined by $x = 5 \cos \theta$, $y = 4 \sin \theta - 3$ at $\theta = \frac{\pi}{6}$.

Then, find if the curve is getting closer or further away from the x -axis and y -axis at this point.

$$\frac{dy}{dx} = \frac{y'}{x'} = \frac{4 \cos \theta}{-5 \sin \theta} = -\frac{4}{5} \cot \theta, \quad \left. \frac{dy}{dx} \right|_{\theta=\pi/6} = \frac{y'(\pi/6)}{x'(\pi/6)} = \boxed{\frac{-4\sqrt{3}}{5}}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{d\theta} \left[\frac{dy}{dx} \right]}{\frac{dx}{d\theta}} = \frac{\frac{4}{5} \csc^2 \theta}{-5 \sin \theta} = -\frac{4}{25} \csc^3 \theta, \quad \left. \frac{d^2y}{dx^2} \right|_{\theta=\pi/6} = \boxed{\frac{-32}{25}}$$

At $\theta = \pi/6$, the curve is at $(\frac{5\sqrt{3}}{2}, -1)$.

"Moving closer" is about change, so look at derivatives. Parametric is great to look at each direction individually

$y'(\pi/6) = 2\sqrt{3} > 0$ so y 's are increasing. Since $y = -1$ and are increasing, the graph gets closer to x -axis

$x'(\pi/6) = -5/2 < 0$ so x 's are decreasing. Since $x = 5\sqrt{3}/2$ and are decreasing, the graph gets closer to y -axis

Arc Length in Parametric

Since both $x(t)$ and $y(t)$ are both functions with respect to t , the proof of arc length in rectangular form is expanded to convert it to parametric equations:

Rectangular $\rightarrow \int_a^b \sqrt{\left(\frac{dx}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + (f'(x))^2} dx$

Parametric $\rightarrow \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

↑
nothing cancels out so this is it!!

Ex. 4: Find the length of the curve whose parametric equation is $x = \cos^3 t$ and $y = \sin^3 t$ from $[0, \frac{\pi}{2}]$.

$$\text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$x' = -3\cos^2 t \sin t$$

$$y' = 3\sin^2 t \cos t$$

$$\int_0^{\pi/2} \sqrt{(-3\cos^2 t \sin t)^2 + (3\sin^2 t \cos t)^2} dt = \int_0^{\pi/2} \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t} dt$$

$$= \int_0^{\pi/2} \sqrt{9\cos^2 t \sin^2 t (\cos^2 t + \sin^2 t)} dt = \int_0^{\pi/2} |3\cos t \sin t| dt$$

Since $\sin t > 0$
and $\cos t > 0$
for $[0, \pi/2]$...

$$= \int_0^{\pi/2} 3\cos t \sin t dt \xrightarrow[\substack{u = \sin t \\ du = \cos t dt}]{u = \sin t} 3 \int_0^1 u du = \frac{3}{2} u^2 \Big|_0^1 = \boxed{\frac{3}{2}}$$

PVA – Related Definitions in Parametric Vector Form

has direction + length

Start → end

Suppose a particle moves along a smooth curve in the plane so that its position at any time t is $\langle x(t), y(t) \rangle$ where x and y are both twice-differentiable functions of t .

- 1) The particle's **position** - $\langle x(t), y(t) \rangle$ each x & y can be found independent of one another
- 2) The particle's **velocity** - $\langle x'(t), y'(t) \rangle$ NOT dy/dx !!
- 3) The particle's **acceleration** - $\langle x''(t), y''(t) \rangle$ NOT $\frac{d^2 y}{dx^2}$ or $\frac{1}{dx} \left[\frac{dy}{dx} \right]$!!
- 4) The particle's **speed** - $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ ← ensures both x & y components of velocity are positive! So it's like a 2-D way to do absolute value!
[magnitude of velocity vector]
- 5) The particle's **distance travelled** - $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ ← Should look familiar... Arc length!
still $\int \text{speed} dt$!!!
- 6) The particle's **displacement** - $\langle \int_a^b \frac{dx}{dt}, \int_a^b \frac{dy}{dt} \rangle \leftarrow \langle x(b) - x(a), y(b) - y(a) \rangle$
still $\int \text{velocity} dt$!
both final-initial!
- 7) The particle's **at rest** when -
when velocity = 0 Both $x' = y' = 0$ ← if only x' or $y' = 0$, then you aren't moving in that 1 direction.... doesn't guarantee at rest

Ex. 4: Find the length of the curve whose parametric equation is $x = \cos^3 t$ and $y = \sin^3 t$ from $[0, \frac{\pi}{2}]$.

$$\text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$x' = -3\cos^2 t \sin t \quad y' = 3\sin^2 t \cos t$$

$$\begin{aligned} & \int_0^{\pi/2} \sqrt{(-3\cos^2 t \sin t)^2 + (3\sin^2 t \cos t)^2} dt = \int_0^{\pi/2} \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t} dt \\ & = \int_0^{\pi/2} \sqrt{9\cos^2 t \sin^2 t (\cos^2 t + \sin^2 t)} dt = \int_0^{\pi/2} |3\cos t \sin t| dt \quad \text{Since } \sin t > 0 \text{ and } \cos t > 0 \text{ for } [0, \pi/2] \dots \\ & = \int_0^{\pi/2} 3\cos t \sin t dt \quad \xrightarrow[\substack{u = \sin t \\ du = \cos t dt}]{u = \sin t} 3 \int_0^1 u du = \frac{3}{2} u^2 \Big|_0^1 = \boxed{\frac{3}{2}} \end{aligned}$$

PVA – Related Definitions in Parametric Vector Form

has direction + length

Start → end

Suppose a particle moves along a smooth curve in the plane so that its position at any time t is $\langle x(t), y(t) \rangle$ where x and y are both twice-differentiable functions of t .

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still $\int \text{speed} dt$!!!
- 6) The particle's **displacement** - $\langle \int_a^b \frac{dx}{dt}, \int_a^b \frac{dy}{dt} \rangle \leftarrow \langle x(b) - x(a), y(b) - y(a) \rangle$
still $\int \text{velocity} dt$!
both final-initial!
- 7) The particle's **at rest** when -
when velocity = 0 Both $x' = y' = 0$ ← if only x' or $y' = 0$, then you aren't moving in that 1 direction.... doesn't guarantee at rest

A graphing calculator is required for problems on this part of the exam.

At time $t \geq 0$, the position of a particle moving along a curve in the xy -plane is $(x(t), y(t))$, where $\frac{dx}{dt} = t - 5 \cos t$ and $\frac{dy}{dt} = 6 \cos(1 + \sin t)$. At time $t = 3$, the particle is at position $(-1, 2)$.

- (a) Write an equation for the line tangent to the path of the particle at time $t = 3$.
- (b) Find the time t when the line tangent to the path of the particle is vertical. Is the direction of motion of the particle up or down at that moment? Give a reason for your answer.
- (c) Find the y -coordinate of the particle's position at time $t = 0$.
- (d) Find the total distance traveled by the particle for $0 \leq t \leq 3$.

a) Still need point & slope! Point: (x, y) at $t = 3 \rightarrow (-1, 2)$

Slope: $\frac{dy}{dx} = \frac{y'}{x'} \Big|_{t=3} = \frac{6 \cos(1 + \sin 3)}{3 - 5 \cos 3} = 0.314 \rightarrow \boxed{y - 2 = 0.314(x + 1)}$

b) Vertical $\rightarrow$ slope is undefined $\rightarrow \frac{dx}{dt} = 0$ at $t = 1.30644$

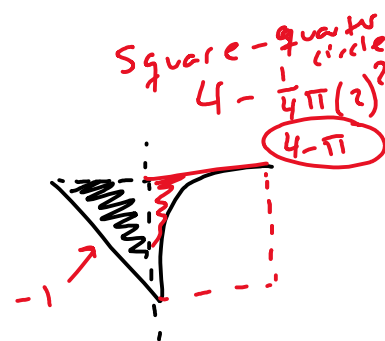
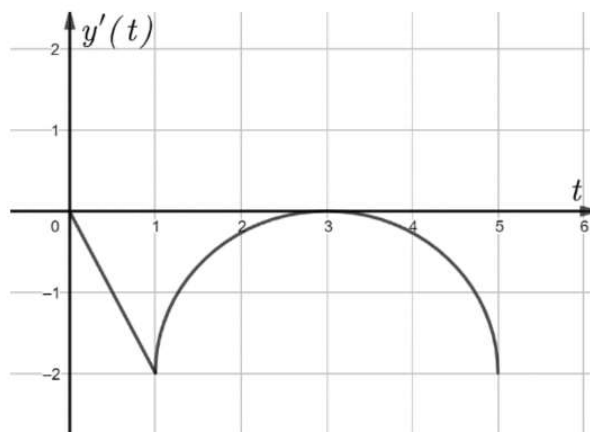
up/down motion is $\frac{dy}{dt} \rightarrow \frac{dy}{dt} \Big|_{t=1.30644} = \boxed{-2.306 < 0}$ so down

c) Given $\frac{dy}{dt} \rightarrow$ find $y \dots$ integrate! But we only care about y 's.

$y(0) = y(3) - \int_0^3 \frac{dy}{dt} dt$, $y(0) = 2 - \int_0^3 6 \cos(1 + \sin t) dt = \boxed{3.634}$
going "back" in time

d) Distance = $\int$ speed $\rightarrow$ in parametric

$$= \int_0^3 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \boxed{13.453}$$



At time t , the position of a particle moving in the xy -plane is given by the parametric functions $(x(t), y(t))$, where $\frac{dx}{dt} = 8te^{2t}$ and $\frac{dy}{dt}$ is shown in the figure above, consisting of a line segment and a semicircle. At time $t = 3$, the particle is at position $(7, -1)$.

- a) Find the position of the particle at time $t = 0$. Given $\frac{dx}{dt}$ & $\frac{dy}{dt} \rightarrow$ Find $x, y \dots$ Integrate!

$$x(0) = x(3) - \int_0^3 8te^{2t} dt \quad \text{Integrate by parts!!}$$

$u = t \quad dv = e^{2t} dt$
 $du = dt \quad v = \frac{1}{2}e^{2t}$

$$x(0) = 7 - 8 \left[\frac{1}{2}te^{2t} - \frac{1}{2} \int_0^3 e^{2t} dt \right]$$

$$x(0) = 7 - 8 \left[\frac{1}{2}te^{2t} - \frac{1}{4}e^{2t} \Big|_0^3 \right] = 7 - 8 \left[\left(\frac{3}{2}e^6 - \frac{1}{4}e^6 \right) - \left(0 - \frac{1}{4} \right) \right] = \boxed{5 - 10e^6}$$

$$y(0) = y(3) - \int_0^3 \frac{dy}{dt} dt \rightarrow y(0) = -1 - \left[\text{area between graph and x-axis from } 0 \rightarrow 3 \right] \rightarrow y(0) = -1 - [-1 - (4 - \pi)]$$

- b) Find the time, if any, when the particle is at rest.

Both $x' = y' = 0$

$$x' = 8te^{2t} = 0$$

$$t = 0$$

← given graph of y'

$$y' = 0$$

$$t = 0, 3$$

so $\boxed{t = 0}$

(not $t = 3$)

- c) Find the acceleration vector at time $t = \frac{1}{2}$.

$\langle x'', y'' \rangle \dots$ Given $\frac{dx}{dt} = x' \rightarrow$ find x'' Differentiate!

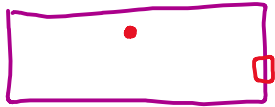
$$x''(t) = 8e^{2t} + 8te^{2t} \cdot 2, \quad x'(\frac{1}{2}) = 8e + 8e = 16e$$

$$y''(t) \text{ is slope of graph of } y'. \quad y''(\frac{1}{2}) = -2$$

acceleration vector =

$$\boxed{\langle 16e, -2 \rangle}$$

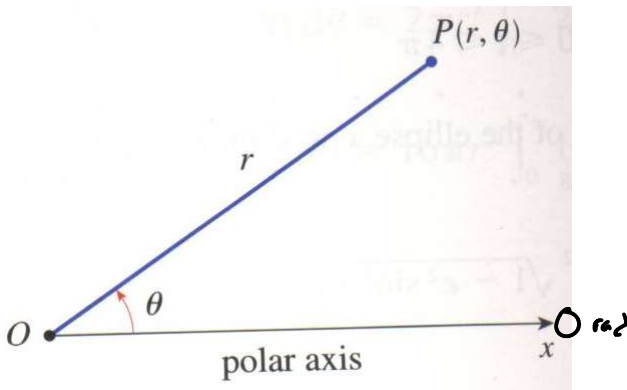
Imagine an empty room and you need to tell someone to get to a door.



AP Calculus II Notes 10.4 Polar Coordinates

- You'd tell them to either
- 1) Go directly right, then go down
RECTANGULAR!!!
 - 2) Rotate some degree and then walk straight
POLAR!!!

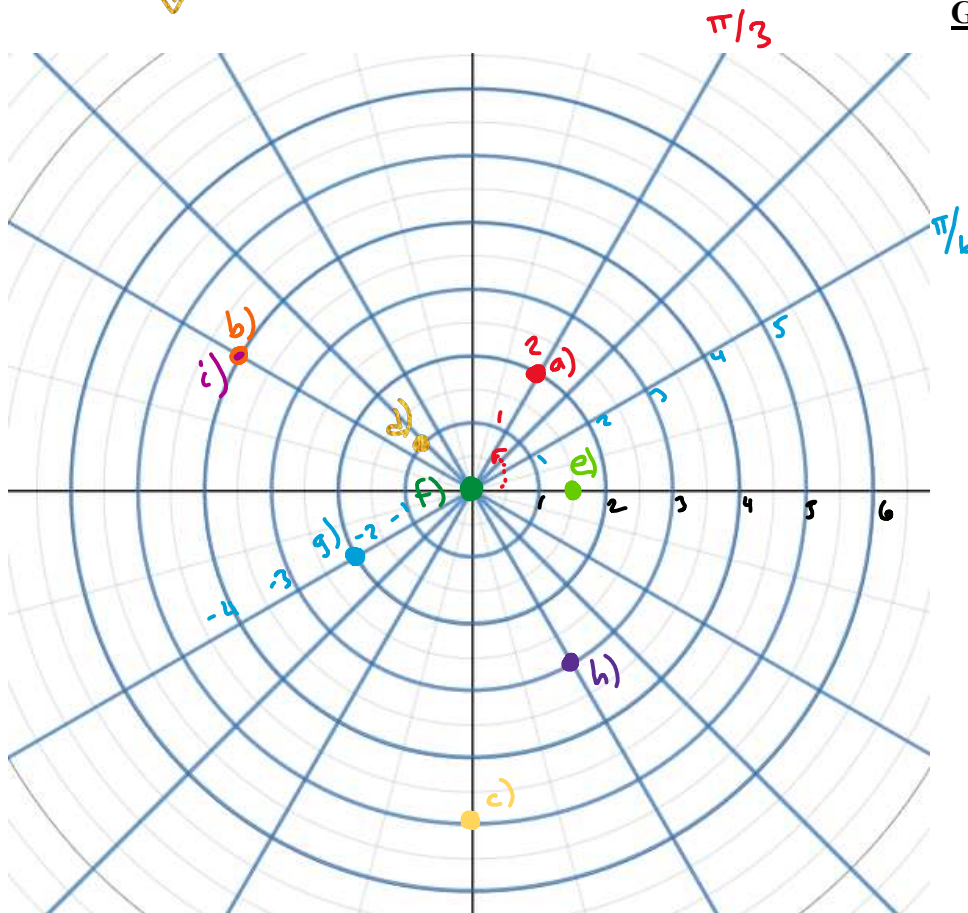
So far, all the graphs you have been working with since middle school are in the rectangular form with points (x, y) . Plotting points always required you to move left or right and then up or down. A new coordinate system is called the **polar coordinate system**.



To form a point in this system, fix a point O , called the **pole** or **origin**, and a ray emanating from O , called the **polar axis**. To locate a point P , in the plane, you draw a segment OP , r is the length of OP and θ is the angle from the polar axis to OP measured in the counterclockwise direction. Polar coordinates are written as (r, θ) .

Instead of move up/down and left/right, this is rotate & emanate

NOTICE: blank grid is concentric circles, not rectangles



Graphing Polar Coordinates Practice:

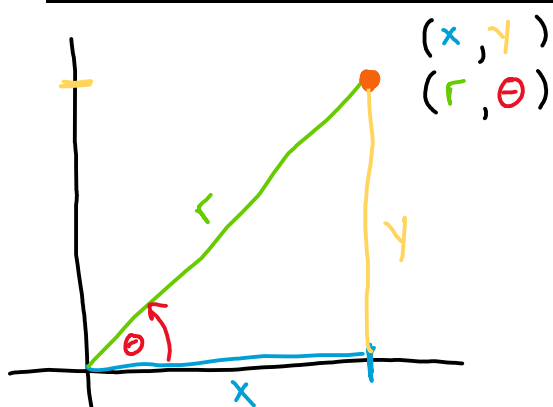
- $\left(2, \frac{\pi}{3}\right)$ Rotate $\pi/3$, emanate out radius of 2
- $\left(4, \frac{5\pi}{6}\right)$ rotate $5\pi/6$, emanate 4
- $\left(5, \frac{3\pi}{2}\right)$
- $\left(1, \frac{3\pi}{4}\right)$
- $(1.5, 0)$ rotate 0, emanate 1.5
- $\left(0, \frac{4\pi}{3}\right)$ rotate $\frac{4\pi}{3}$, emanate 0 (at pole)
- $\left(-2, \frac{\pi}{6}\right)$ rotate $\pi/6$, negative r means go through pole to other side
- $\left(3, -\frac{\pi}{3}\right)$ rotate $-\pi/3$ means go clockwise
- $\left(-4, \frac{11\pi}{6}\right)$ at same location as b)

these are ∞ ways to end at same location....

add or subtract 2π to get coterminal angles...

r could be $\pm$

Rectangular $\longleftrightarrow$ Polar Conversion:



$$\cos \theta = \frac{x}{r}$$

$$\sin \theta = \frac{y}{r}$$

$$\tan \theta = \frac{y}{x}$$

$$x^2 + y^2 = r^2$$

polar $\rightarrow$ rectangular

$$\boxed{\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}}$$

for each (r, θ) there is 1 (x, y) !

rectangular $\rightarrow$ polar

∞ θ -values to $\tan \theta = \frac{y}{x}$
2 r -values to $x^2 + y^2 = r^2$

so for each (x, y) , there are ∞ many (r, θ) !

Ex. 1: Convert the following polar coordinates into rectangular coordinates:

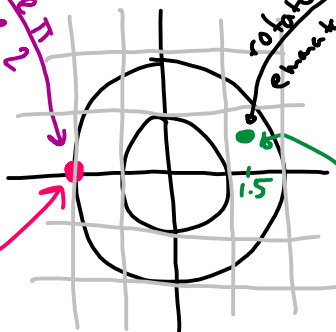
a) $(r, \theta) = (2, \pi)$

$$x = r \cos \theta = 2 \cos \pi = -2$$

$$y = r \sin \theta = 2 \sin \pi = 0$$

$$(x, y) = (-2, 0)$$

rotate π
clockwise 2



b) $(r, \theta) = (\sqrt{3}, \frac{\pi}{6}) = (1.732, \pi/6)$

$$x = r \cos \theta = \sqrt{3} \cos \pi/6 = \sqrt{3}(\frac{\sqrt{3}}{2}) = \frac{3}{2}$$

$$y = r \sin \theta = \sqrt{3} \sin \pi/6 = \sqrt{3}(\frac{1}{2}) = \frac{\sqrt{3}}{2}$$

$$(x, y) = (\frac{3}{2}, \frac{\sqrt{3}}{2}) = (1.5, 0.866)$$

rotate $\pi/6$
counter-clockwise 1.732

Ex. 2: Convert the following rectangular coordinates into 2 sets of polar coordinates:

a) $(x, y) = (-1, 1)$

$$\tan \theta = \frac{y}{x} = \frac{1}{-1} = -1, \tan \theta = -1 \text{ at } \theta = -\pi/4, 3\pi/4, 7\pi/4, \dots$$

$\theta = -\pi/4$: use $x =$ or $y =$ to find r

$$x = r \cos \theta$$

$$-1 = r \cos -\pi/4$$

$$-1 = r(\frac{1}{\sqrt{2}})$$

$$r = -\sqrt{2}$$

$$y = r \sin \theta$$

$$1 = r \sin -\pi/4$$

$$1 = r(-\frac{1}{\sqrt{2}})$$

$$r = -\sqrt{2}$$

$$\boxed{(-\sqrt{2}, -\pi/4)} \text{ or } \boxed{(-\sqrt{2}, 7\pi/4)}$$

$$\theta = 3\pi/4: -1 = r \cos 3\pi/4$$

$$-1 = r(-\frac{1}{\sqrt{2}}), r = \sqrt{2}$$

$$\boxed{(\sqrt{2}, 3\pi/4)}$$

b) $(x, y) = (0, 4)$

$$\tan \theta = \frac{y}{x}, \tan \theta \rightarrow \frac{4}{0} \text{ is undefined at } \theta = \pi/2, 3\pi/2, \dots$$

$$\theta = \pi/2: x = r \cos \theta, 0 = r \cos \pi/2$$

$$0 = r \cdot 0 \dots \text{useless}$$

$$y = r \sin \theta \rightarrow 4 = r \sin \pi/2$$

$$4 = r \cdot 1, r = 4$$

$$\boxed{(4, \pi/2)}$$

$\leftarrow$ makes sense when you envision plotting this

$$\theta = 3\pi/2: 4 = r \sin 3\pi/2,$$

$$4 = r \cdot (-1), r = -4$$

$$\boxed{(-4, 3\pi/2)}$$

rectangular: $y = f(x)$ as x increases (go right), how does y change (move up/down)

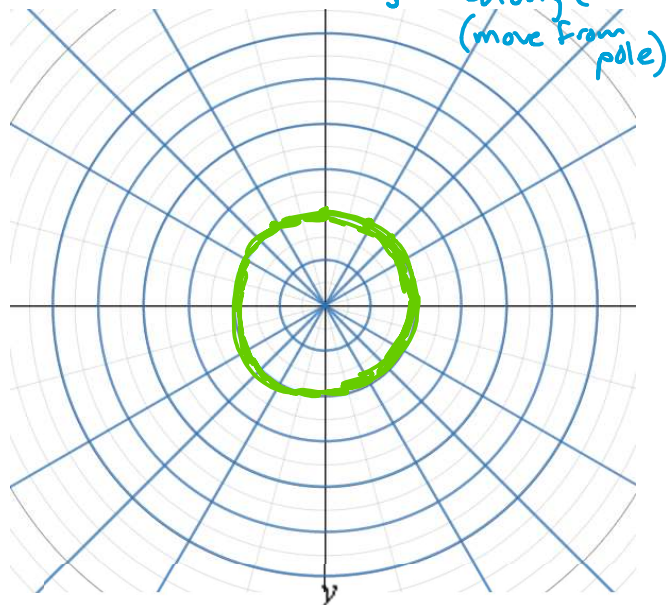
Graphing Polar Equations: $r = f(\theta)$ as θ increases (rotate) how does r change (move from pole)

Ex. 3: Graph the following polar equations:

a) $r = 2$

θ	r
0	2
$\pi/6$	2
$\pi/4$	2
$\pi/3$	2
$\pi/2$	2
π	2
2π	2

$r = c$ is
circle of
radius c



b) $\theta = \frac{\pi}{3}$

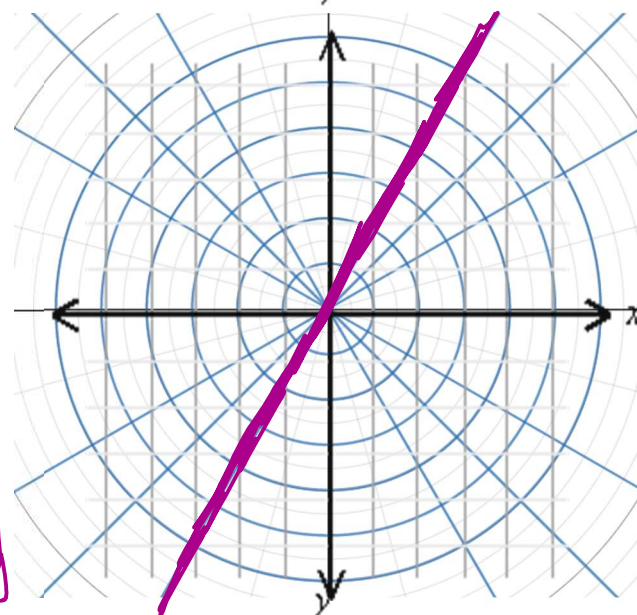
$\theta = c$ is a line

$\tan \theta = \frac{y}{x}$

$y = x \tan \theta$

$y = x \tan \pi/3 \rightarrow$ rectangular $y = \sqrt{3} \cdot x + 0$

$m = \sqrt{3}, b = 0$



c) $r = 2 \csc \theta$

θ | $r = 2 \csc \theta = \frac{2}{\sin \theta}$

0 | $r = \frac{2}{\sin 0}$... undefined??

$\pi/6$ | $r = \frac{2}{\sin \pi/6} = \frac{2}{1/2} = 4$

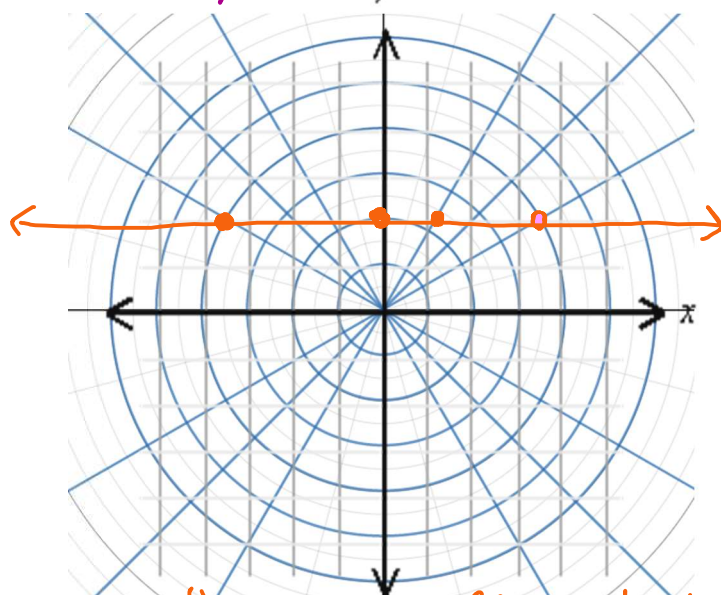
$\pi/3$ | $r = \frac{2}{\sin \pi/3} = \frac{2}{\sqrt{3}/2} = \frac{4}{\sqrt{3}} \approx 2.3$

$\pi/2$ | $r = \frac{2}{\sin \pi/2} = \frac{2}{1} = 2$

$5\pi/6$ | $r = \frac{2}{1/2} = 4$

π | $r = \frac{2}{\sin \pi} \rightarrow$ undefined??

$7\pi/6$ | $r = \frac{2}{-1/2} = -4 \leftarrow$ overlapping itself



Horizontal line that
never reaches $\theta = 0$ or π !
 $r = \frac{2}{\sin \theta}, r \sin \theta = 2 \rightarrow y = 2$

d) Limaçon: $r = 2 - 4 \sin \theta$

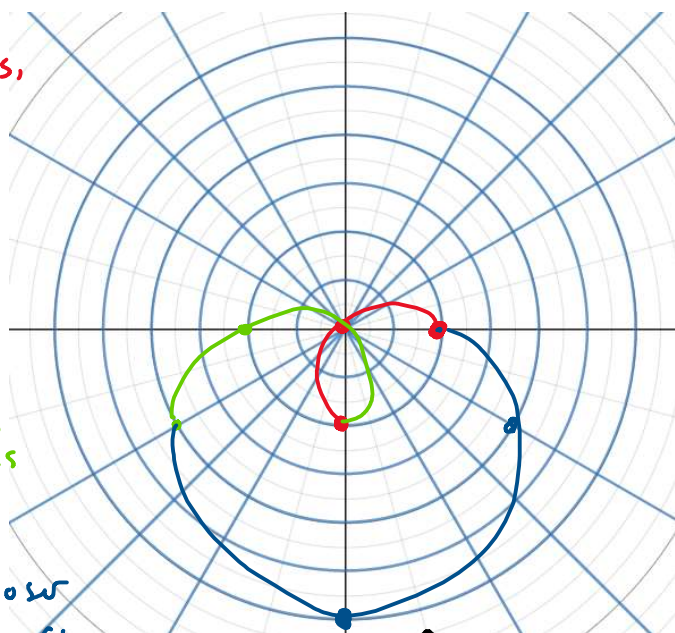
to make graphing easier, let's use the $\pi/6$'s because $\sin \pi/6 = 1/2$ (not $1/2$ or $5/2 \dots$)

θ	$r = 2 - 4 \sin \theta$
0	$2 - 4 \sin 0 = 2$
$\pi/6$	$2 - 4(1/2) = 0$
$\pi/2$	$2 - 4(1) = -2$
$5\pi/6$	$2 - 4(1/2) = 0$
π	$2 - 4(0) = 2$
$7\pi/6$	$2 - 4(-1/2) = 4$
$3\pi/2$	$2 - 4(-1) = 6$
$11\pi/6$	$2 - 4(-1/2) = 4$
2π	$2 - 4(0) = 2$

as θ increases, r decreases from 2 to -2

returns to pole and then gets further as θ increases

gets closer to pole as radius decreases as θ increases

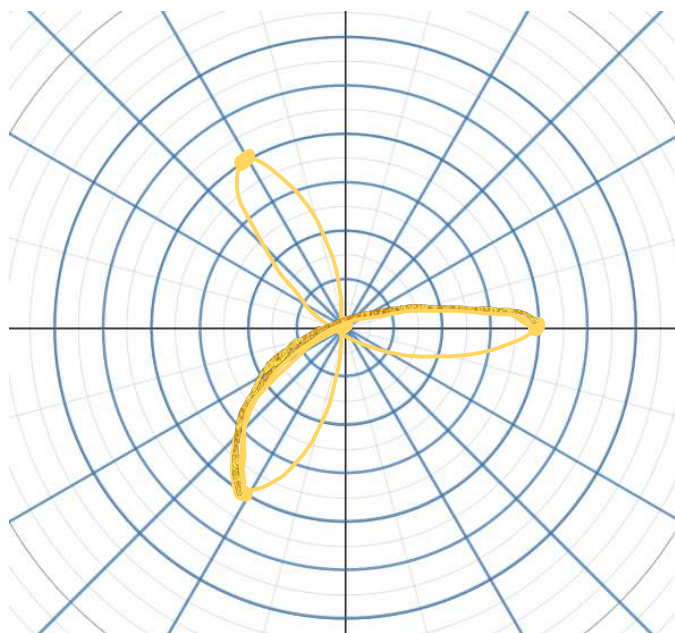


↑
Limaçon with inner loop

e) Rose: $r = 4 \cos 3\theta$

θ	$r = 4 \cos 3\theta$
0	$r = 4 \cos 0 = 4$
$\pi/6$	$r = 4 \cos \pi/2 = 0$
$\pi/3$	$r = 4 \cos \pi = -4$
$\pi/2$	$r = 4 \cos 3\pi/2 = 0$
$2\pi/3$	$r = 4 \cos 2\pi = 4$
$5\pi/6$	$r = 4 \cos 5\pi/2 = 0$
π	$r = 4 \cos 3\pi = -4$
$7\pi/6$	$r = 4 \cos 7\pi/2 = 0$
$4\pi/3$	$r = 4 \cos 4\pi = 4$

Starts to retrace itself, so we can stop



d) Limaçon: $r = 2 - 4 \sin \theta$

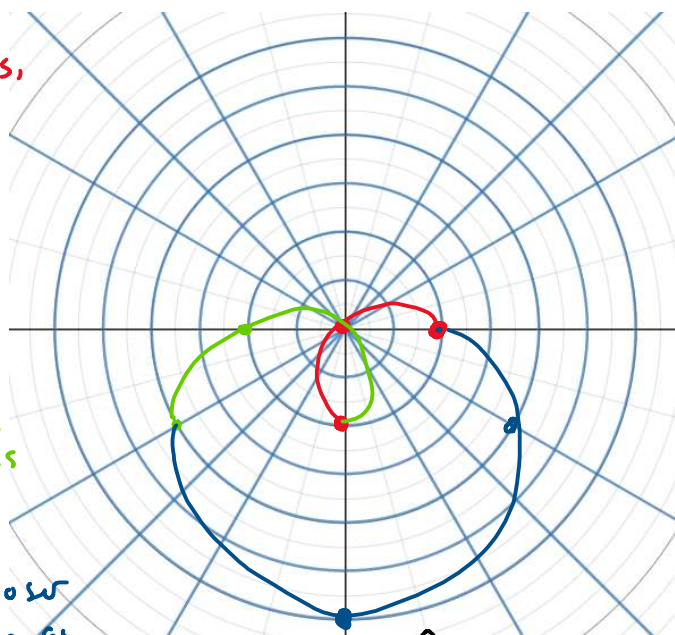
to make graphing easier, let's use the $\pi/6$'s because $\sin \pi/6 = 1/2$ (not $1/2$ or $5/2 \dots$)

θ	$r = 2 - 4 \sin \theta$
0	$2 - 4 \sin 0 = 2$
$\pi/6$	$2 - 4(1/2) = 0$
$\pi/2$	$2 - 4(1) = -2$
$5\pi/6$	$2 - 4(1/2) = 0$
π	$2 - 4(0) = 2$
$7\pi/6$	$2 - 4(-1/2) = 4$
$3\pi/2$	$2 - 4(-1) = 6$
$11\pi/6$	$2 - 4(-1/2) = 4$
2π	$2 - 4(0) = 2$

as θ increases, r decreases from 2 to -2

returns to pole and then gets further as θ increases

gets closer to pole as radius decreases as θ increases

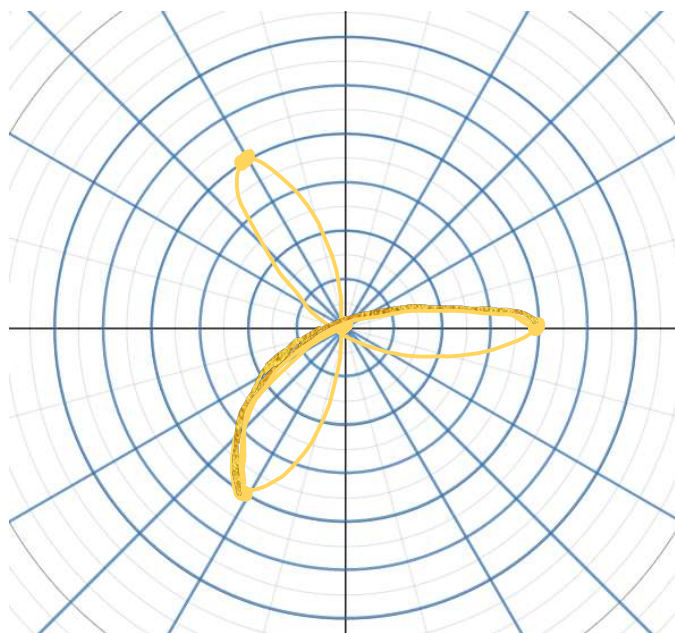


↑
Limaçon with inner loop

e) Rose: $r = 4 \cos 3\theta$

θ	$r = 4 \cos 3\theta$
0	$r = 4 \cos 0 = 4$
$\pi/6$	$r = 4 \cos \pi/2 = 0$
$\pi/3$	$r = 4 \cos \pi = -4$
$\pi/2$	$r = 4 \cos 3\pi/2 = 0$
$2\pi/3$	$r = 4 \cos 2\pi = 4$
$5\pi/6$	$r = 4 \cos 5\pi/2 = 0$
π	$r = 4 \cos 3\pi = -4$
$7\pi/6$	$r = 4 \cos 7\pi/2 = 0$
$4\pi/3$	$r = 4 \cos 4\pi = 4$

Starts to retrace itself, so we can stop



Polar Coordinates and Calculus

If f is a differentiable function of θ , then the *slope* of the tangent line to the graph of $r = f(\theta)$ at the point (r, θ) is found the same way as a parametric equation. In fact, we will convert polar equations to parametric equations using the relationships:

$$\begin{aligned} * x &= r \cos \theta \\ * y &= r \sin \theta \end{aligned}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \leftarrow \begin{array}{l} \text{generally will be product} \\ \text{rules for each!} \end{array}$$

Ex. 4: Find $\frac{dr}{d\theta}$, $\frac{dy}{d\theta}$, $\frac{dx}{d\theta}$ and the slope of the tangent line to $r = 4 \cos \theta$ at $\theta = \frac{\pi}{4}$ and then, explain the meaning of each.

$$\frac{dr}{d\theta} = -4 \sin \theta, \quad \left. \frac{dr}{d\theta} \right|_{\theta=\pi/4} = -4\left(\frac{\sqrt{2}}{2}\right) = \boxed{-2\sqrt{2}}$$

Since $\frac{dr}{d\theta} < 0$, the curve's radius is decreasing... aka gets closer to the pole!

$$y = 4 \cos \theta \sin \theta \rightarrow \frac{dy}{d\theta} = -4 \sin^2 \theta + 4 \cos^2 \theta, \quad \left. \frac{dy}{d\theta} \right|_{\theta=\pi/4} = -4\left(\frac{\sqrt{2}}{2}\right)^2 + 4\left(\frac{\sqrt{2}}{2}\right)^2 = \boxed{0}$$

Since $\frac{dy}{d\theta} = 0$, the y-values aren't changing... aka the graph isn't moving up or down

$$x = 4 \cos^2 \theta \rightarrow \frac{dx}{d\theta} = -8 \cos \theta \sin \theta$$

$$\left. \frac{dx}{d\theta} \right|_{\theta=\pi/4} = -8\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \boxed{-4}$$

Since $\frac{dx}{d\theta} < 0$, the x-values are decreasing... aka the curve is moving left!

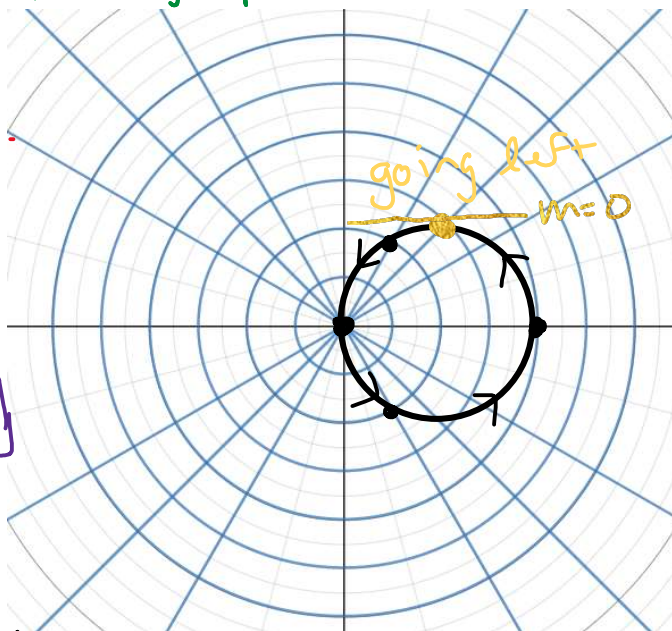
Slope of the tangent line = $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-4 \sin^2 \theta + 4 \cos^2 \theta}{-8 \cos \theta \sin \theta} \bigg|_{\theta=\pi/4} = \frac{0}{-4} = \boxed{0}$$

there is a horizontal tangent line at $\theta = \pi/4$.

Horizontal tangent lines occur when $-\frac{dy}{d\theta} = 0$, $\frac{dx}{d\theta} \neq 0$

Vertical tangent lines occur when $-\frac{dx}{d\theta} = 0$, $\frac{dy}{d\theta} \neq 0$



θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
r	4	2	0	-2	-4

* know

$$r = a \cos \theta$$

$$+ r = a \sin \theta$$

are circles of diameter a

Ex. 5: Find all points (r, θ) of horizontal and vertical tangency to $r = 2(1 - \cos \theta)$ from $0 \leq \theta < 2\pi$.

$$\frac{dy}{d\theta} = 0 \quad \frac{dx}{d\theta} = 0$$

Horizontal: need $\frac{dy}{d\theta}$, but given $r = \dots$ $y = r \sin \theta$, then differentiate!

$$y = (2 - 2\cos \theta) \sin \theta, \quad \frac{dy}{d\theta} = 2\sin^2 \theta + (2 - 2\cos \theta) \cos \theta = 0$$

$$\text{at } \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\text{So points } (r, \theta) = (\cancel{0}, 0), (3, \frac{2\pi}{3}), (3, \frac{4\pi}{3})$$

$$\text{Vertical: } x = (2 - 2\cos \theta) \cos \theta = 2\cos \theta - 2\cos^2 \theta$$

$$\frac{dx}{d\theta} = -2\sin \theta + 4\cos \theta \sin \theta = 0$$

$$= 2\sin \theta (2\cos \theta - 1) = 0$$

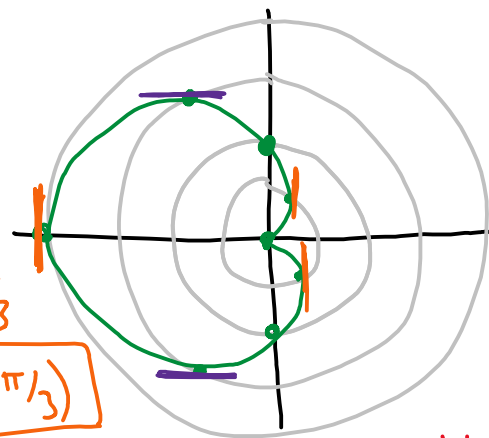
$$2\sin \theta = 0$$

$$\theta = 0, \pi$$

$$2\cos \theta - 1 = 0$$

$$\cos \theta = \frac{1}{2} \text{ at } \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\text{So points } (r, \theta) = (\cancel{0}, 0), (4, \pi), (1, \frac{\pi}{3}), (1, \frac{5\pi}{3})$$



Throw out $(0,0)!!$

Ex. 6: Find the equation of the tangent line to $r = \sin \theta$ at $\theta = \frac{\pi}{3}$.

Point (x, y) and slope $\frac{dy}{dx}$

$$x = \sin \theta \cos \theta, \quad \frac{dx}{d\theta} = \cos^2 \theta - \sin^2 \theta$$

$$x(\frac{\pi}{3}) = (\frac{\sqrt{3}}{2})(\frac{1}{2}) = \frac{\sqrt{3}}{4}, \quad \frac{dx}{d\theta} \Big|_{\theta=\frac{\pi}{3}} = (\frac{1}{2})^2 - (\frac{\sqrt{3}}{2})^2 = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$$

$$y = \sin^2 \theta, \quad \frac{dy}{d\theta} = 2\sin \theta \cos \theta$$

$$y(\frac{\pi}{3}) = (\frac{\sqrt{3}}{2})^2 = \frac{3}{4}, \quad \frac{dy}{d\theta} \Big|_{\theta=\frac{\pi}{3}} = 2(\frac{\sqrt{3}}{2})(\frac{1}{2}) = \frac{\sqrt{3}}{2}$$

$$\text{point: } (\frac{\sqrt{3}}{4}, \frac{3}{4}) \quad \text{slope: } \frac{dy}{dx} = \frac{y'}{x'} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$$

$$\text{equation: } y - \frac{3}{4} = -\sqrt{3}(x - \frac{\sqrt{3}}{4})$$

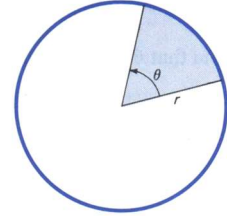
AP Calculus II
Notes 10.5
Polar Area and Arc Length

The development of the polar area formula is similar to that in the rectangular coordinate system. Instead of using rectangles, this system uses sectors of a circle as the basic elements that are being added up.

Proof Using Sectors of a Circle –

In Geometry, we learned that the area of a sector of a circle is $\frac{n^\circ}{360^\circ}(\pi r^2)$. Since we use radians in Calculus, we will translate that to $\frac{\theta}{2\pi}(\pi r^2)$ where θ is the fraction of the circle or angle and $r = f(\theta)$.

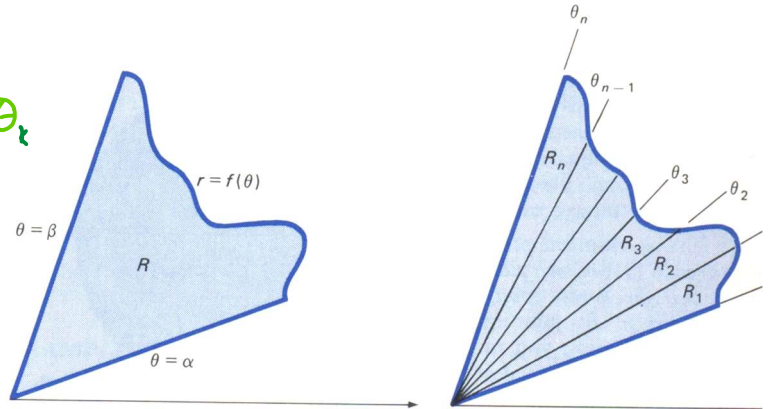
$$\text{Area of sector} = \frac{1}{2} \theta r^2 = \frac{1}{2} (f(\theta))^2 \theta$$



Just like in rectangular mode, where area is infinitely many rectangles added together, polar area has infinitely many sectors added together. So, the area of a region bounded by $\theta = a$, $\theta = b$ and $r = f(\theta)$ where f is continuous and positive and $a \leq \theta \leq b$ is...

$$\text{Area of region} \approx \sum_{k=1}^n \frac{1}{2} r^2 \theta_k$$

to get exact area, let's add ∞ many sectors that have a $\theta \rightarrow 0$



$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{2} r^2 \theta_k = \int_a^b \frac{1}{2} r^2 d\theta$$

$d\theta$ is fraction of circle that approaches 0

Area of a Region in Polar

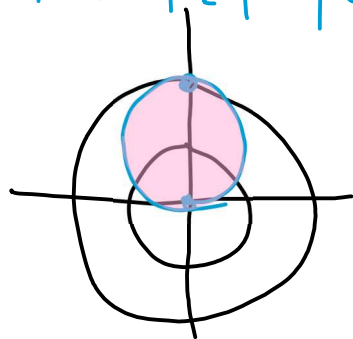
$$A = \frac{1}{2} \int_a^b r^2 d\theta$$

* not volume,
 no π out
 front *

Ex. 1: Find the area enclosed by $r = 2 \sin \theta$.

Area = $\int_a^b \frac{1}{2} r^2 d\theta$... we don't know the bounds to enclose r .

From $[0, \pi]$

$$\begin{array}{c|c|c|c|c|c} \theta & 0 & \pi/6 & \pi/2 & 5\pi/6 & \pi \\ \hline r & 0 & 1 & 2 & 1 & 0 \end{array}$$


$$\text{Area} = \frac{1}{2} \int_0^{\pi} (2 \sin \theta)^2 d\theta$$

$$= 2 \int_0^{\pi} \sin^2 \theta d\theta = 2 \int_0^{\pi} \left[\frac{1}{2} - \frac{1}{2} \cos 2\theta \right] d\theta$$

$$= \int_0^{\pi} (1 - \cos 2\theta) d\theta = \int_0^{\pi} 1 d\theta - \frac{1}{2} \int_0^{\pi} \cos 2\theta d\theta$$

$$= \theta \Big|_0^{\pi} - \frac{1}{2} \sin u \Big|_0^{2\pi} = (\pi - 0) - (0 - 0) = \boxed{\pi}$$

"power reducing"

Identities you MUST know:

$$\ast \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\ast \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

Ex. 2: Find the area of $r = 2 + 2 \cos \theta$ below the polar axis, as shown in the figure.

plugging in some θ 's, we see the figure is made counterclockwise

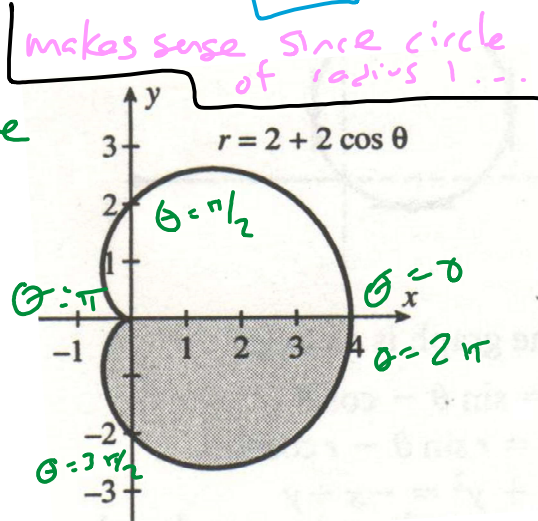
$$\text{Area} = \frac{1}{2} \int_a^b r^2 d\theta$$

$$\text{Area} = \frac{1}{2} \int_{\pi}^{2\pi} (2 + 2 \cos \theta)^2 d\theta$$

$$= \frac{1}{2} \int_{\pi}^{2\pi} (4 + 8 \cos \theta + 4 \cos^2 \theta) d\theta = \int_{\pi}^{2\pi} (2 + 4 \cos \theta) d\theta + 2 \int_{\pi}^{2\pi} \cos^2 \theta d\theta$$

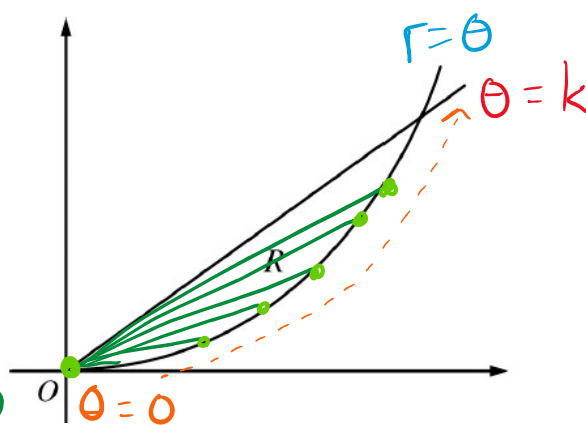
$$= \int_{\pi}^{2\pi} (2 + 4 \cos \theta) d\theta + 2 \int_{\pi}^{2\pi} \left(\frac{1}{2} + \frac{1}{2} \cos 2\theta \right) d\theta$$

$$= \int_{\pi}^{2\pi} (3 + 4 \cos \theta) d\theta + \frac{1}{2} \int_{\pi}^{2\pi} \cos 2\theta d\theta = 3\theta + 4 \sin \theta + \frac{1}{2} \sin 2\theta \Big|_{\pi}^{2\pi} = 6\pi - 3\pi = \boxed{3\pi}$$



Ex. 3:

accumulate all thin sectors shown going from $r=0$ to $r=\theta$



* don't think of top-bottom since polar isn't up/down*

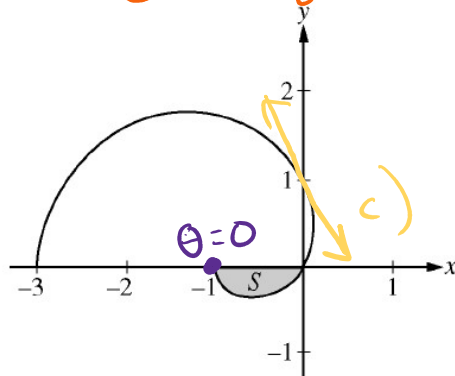
Let R be the region in the first quadrant that is bounded by the polar curves $r = \theta$ and $\theta = k$, where k is a constant, $0 \leq k \leq \frac{\pi}{2}$, as shown in the figure. What is the area of R in terms of k ?

$A = \frac{1}{2} \int_a^b r^2 d\theta \rightarrow$ as θ goes from $a \rightarrow b$, area is between curve and pole!

$A = \frac{1}{2} \int_0^k \theta^2 d\theta = \frac{1}{6} \theta^3 \Big|_0^k = \boxed{\frac{1}{6} k^3}$

Ex. 4:

CA



θ	r
0	-1
$\pi/2$	1
π	3

The graph of the polar curve $r = 1 - 2 \cos \theta$ for $0 \leq \theta \leq \pi$ is shown above. Let S be the shaded region in the third quadrant bounded by the curve and the x -axis.

- Write an integral expression for the area of S .
- Write expression for $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ .
- Write an equation in terms of x and y for the line tangent to the graph of the polar curve at the point where $\theta = \frac{\pi}{2}$. Show the computations that lead to your answer.

a) $A = \frac{1}{2} \int_a^b r^2 d\theta$ bounds are from $\theta = 0$ to where $r = 0$ (at pole) $0 = 1 - 2 \cos \theta$, $\cos \theta = 1/2$, $\theta = \pi/3$

$= \frac{1}{2} \int_0^{\pi/3} (1 - 2 \cos \theta)^2 d\theta = \boxed{0.272}$

b) $x = (1 - 2 \cos \theta) \cos \theta$

$dx/d\theta = (2 \sin \theta) \cos \theta - (1 - 2 \cos \theta) \sin \theta$

$y = (1 - 2 \cos \theta) \sin \theta$

$dy/d\theta = 2 \sin \theta \sin \theta + (1 - 2 \cos \theta) \cos \theta$

c) pt: $(x(\pi/2), y(\pi/2)) = (0, 1)$

slope = $\frac{y'}{x'} = \frac{2+0}{0-1} = -2$

$y - 1 = -2(x - 0)$

CA

Since r is continuous, r must pass through 0 by IVT twice

Ex. 5: Find the area in the inner loop of $r = 2 - 4 \sin \theta$.

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	2	-2	2	6	2

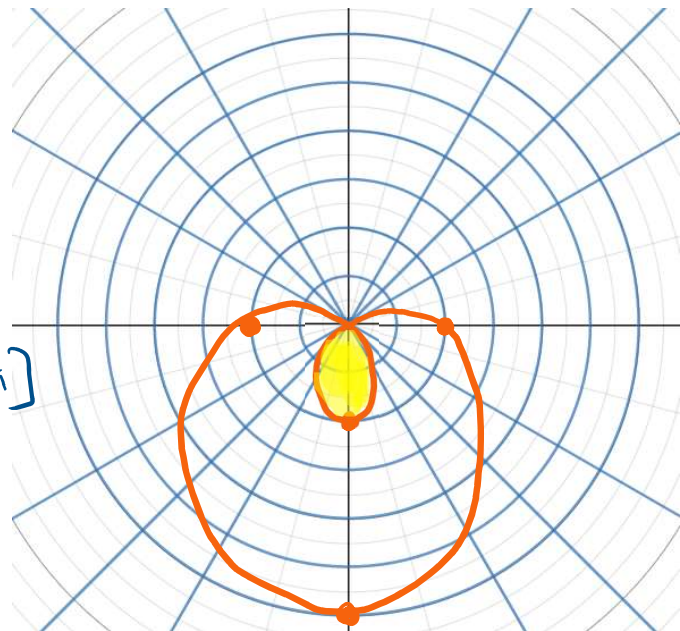
$$\text{Area} = \frac{1}{2} \int_a^b r^2 d\theta$$

loop begins + ends with $r=0$ as it passes through pole

$$0 = 2 - 4 \sin \theta, \sin \theta = 1/2, \theta = \pi/6, 5\pi/6$$

between $[0, \pi/2]$ $[\pi/2, \pi]$

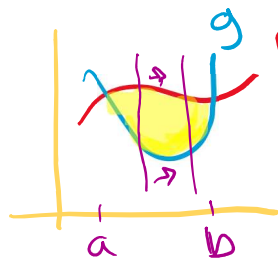
$$\frac{1}{2} \int_{\pi/6}^{5\pi/6} (2 - 4 \sin \theta)^2 d\theta = \boxed{2.174}$$



Area Between Two Curves

Ex. 6: Find the area of the region inside the polar curve $r = 4 \sin \theta$ and outside the polar curve $r = 2$.

A quick sketch will be helpful! For area between curves, think of what we did in rectangular

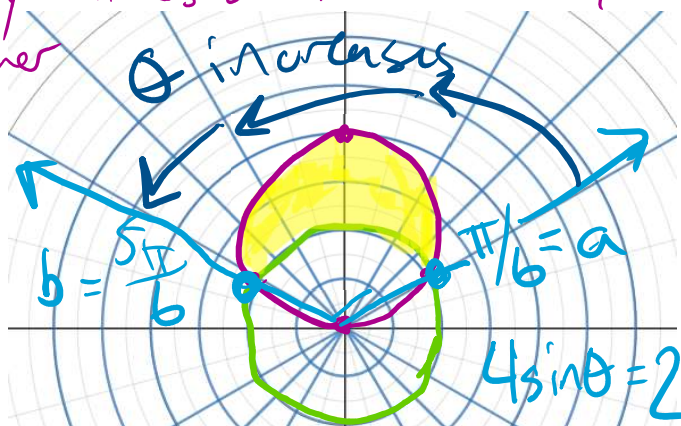


As x increases from $a \rightarrow b$, the graph moves right. At each x -value, the y -values of f are always greater than g (further from x -axis). So

$$\int_a^b f(x) dx - \int_a^b g(x) dx$$

Area of f - Area of g

Polar is same concept! As θ goes from $a \rightarrow b$, the graph rotates. Then we see which r -values are greater (further from pole)



$$A = \frac{1}{2} \int_{\pi/6}^{5\pi/6} [(4 \sin \theta)^2 - (2)^2] d\theta$$

$4 \sin \theta = 2$
 $\sin \theta = 1/2$
 $\theta = \pi/6, 5\pi/6$

$= 7.653$

Ex. 7: Find the area of the region inside the curve $r = 4 \cos \theta$ and outside the curve $r = 2 \cos \theta$.

To make these a little easier, try to take advantage of symmetry! Here, use $\theta = 0$.

Then, the region ends its symmetry when both curves reach the pole. So,

$$4 \cos \theta = 2 \cos \theta = 0, \cos \theta = 0 \\ \theta = \pi/2$$

From $\theta = 0$ to $\theta = \pi/2$, $4 \cos \theta$ is always further from the pole!

$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/2} [(4 \cos \theta)^2 - (2 \cos \theta)^2] d\theta = \boxed{9.425}$$

Ex. 8: Setup, but do not evaluate, the area of the region described.

a) Common interior of $r = 4 \cos \theta$ and $r = 2$.

There is symmetry about $\theta = 0$. The symmetry ends when $r = 4 \cos \theta = 0$, or at $\theta = \pi/2$. As θ goes from $0 \rightarrow \pi/2$, the function furthest from the pole changes

From $\theta = 0$ to the intersection,

$r = 2$ makes up the region and goes straight to the pole.

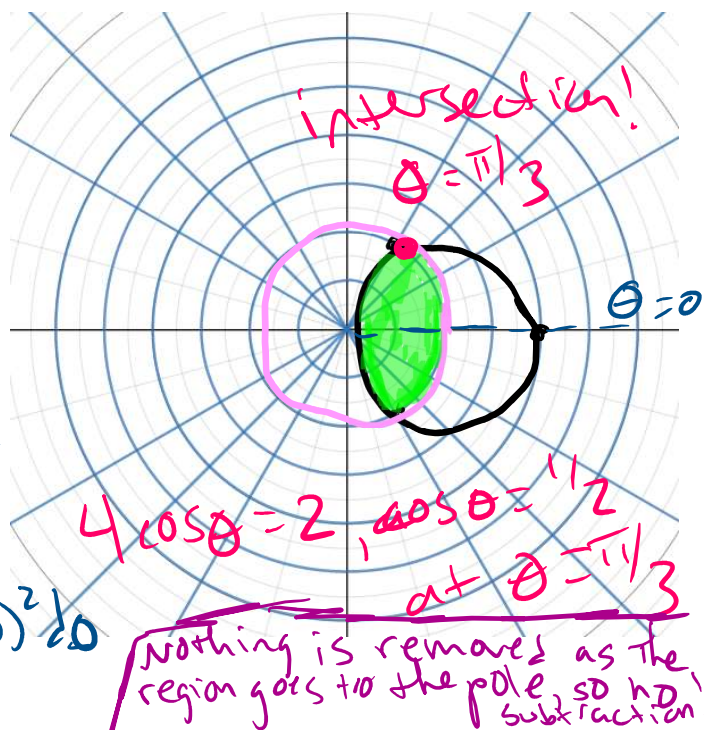
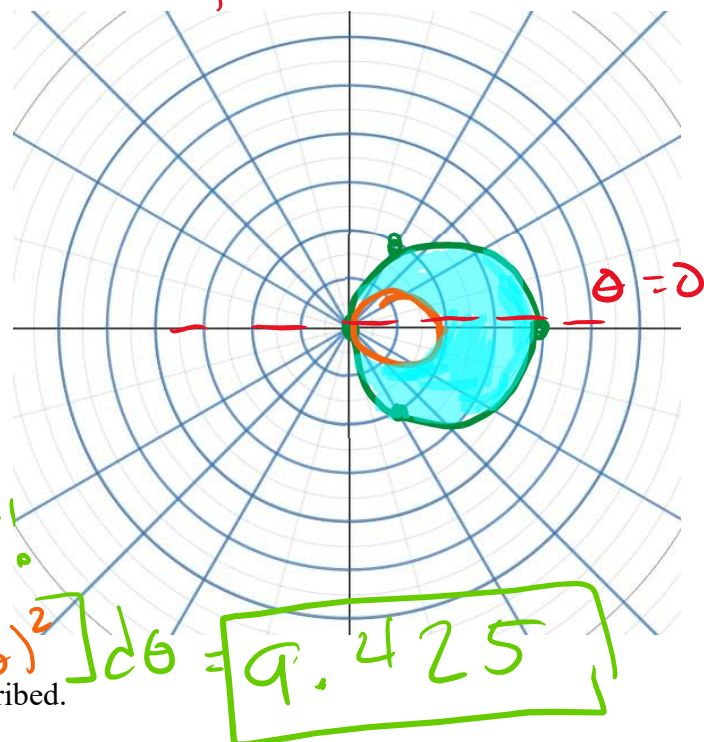
From intersection to $\theta = \pi/2$,

$r = 4 \cos \theta$ is used in the

region. So we add 2 areas

$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/3} (2)^2 d\theta + 2 \cdot \frac{1}{2} \int_{\pi/3}^{\pi/2} (4 \cos \theta)^2 d\theta$$

← symmetry →



b) Common interior of $r = 3 \cos \theta$ and $r = 1 + \cos \theta$.

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	2	1	0	1	2

Symmetry ends when $3 \cos \theta = 0$
at $\theta = \pi/2$.

From $\theta = 0$ to $\theta = \pi/2$, 2 different curves make-up the outside.

The regions switch the outside

From $\theta = 0$ to $\theta = \pi/3$, $r = 1 + \cos \theta$ is used. From $\theta = \pi/3$ to $\theta = \pi/2$

$r = 3 \cos \theta$ is used (small sliver)

$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/3} (1 + \cos \theta)^2 d\theta + 2 \cdot \frac{1}{2} \int_{\pi/3}^{\pi/2} (3 \cos \theta)^2 d\theta$$

c) Inside $r = 2$ and outside $r = 2 - 2 \sin \theta$. Then, determine at what rate the distance between the two functions is changing when $\theta = \frac{2\pi}{3}$.

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	2	0	2	4	2

Symmetry starts when $2 = 2 - 2 \sin \theta$, $\sin \theta = 0$, $\theta = 0$ and ends at $\theta = \pi/2$. From $\theta = 0$ to $\theta = \pi/2$

$r = 2$ is always further from the pole than

$r = 2 - 2 \sin \theta$. This means

area is removed, so subtraction!

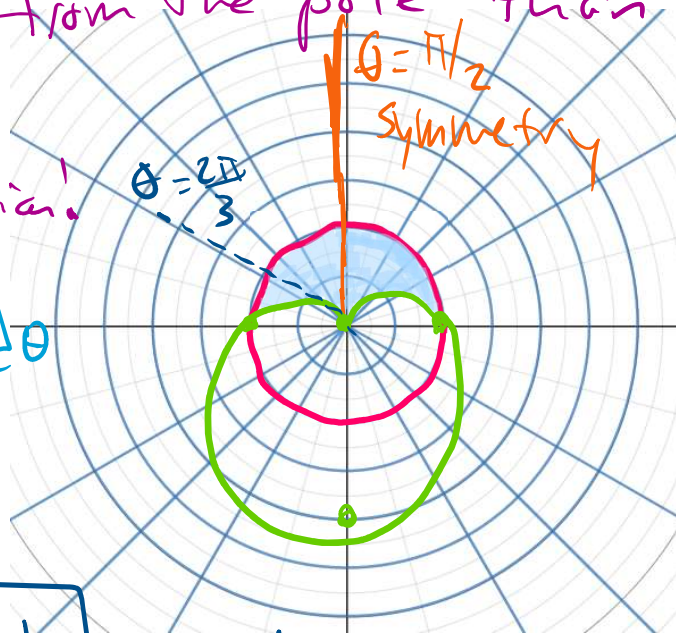
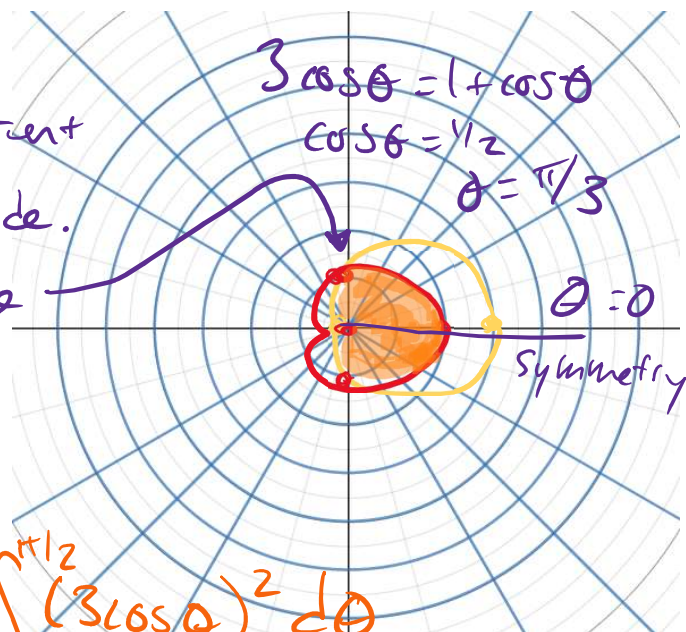
$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/2} [(2)^2 - (2 - 2 \sin \theta)^2] d\theta$$

Distance between curves is

$$D = 2 - (2 - 2 \sin \theta) = 2 \sin \theta$$

$$\text{Change} = \frac{dD}{d\theta} = 2 \cos \theta \Big|_{\theta = 2\pi/3} = \boxed{-1}$$

so the curves are getting closer

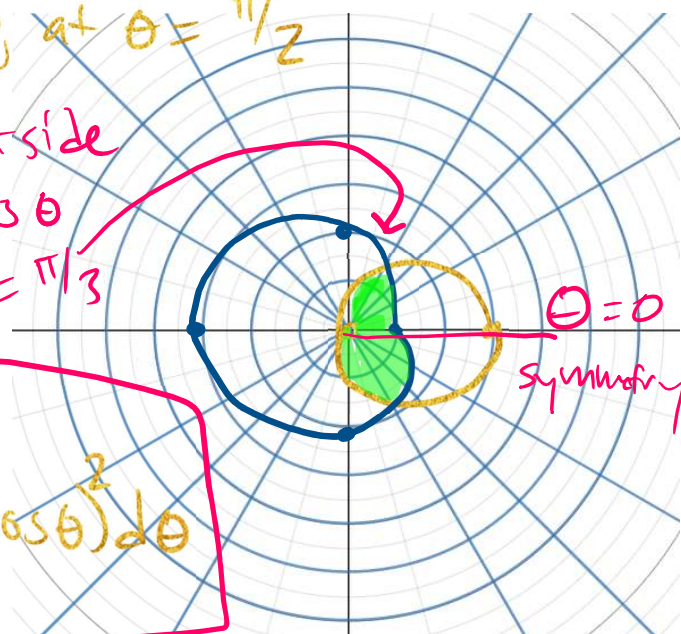


d) Enclosed by $r = 3 \cos \theta$ and $r = 2 - \cos \theta$.

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	1	2	3	2	1

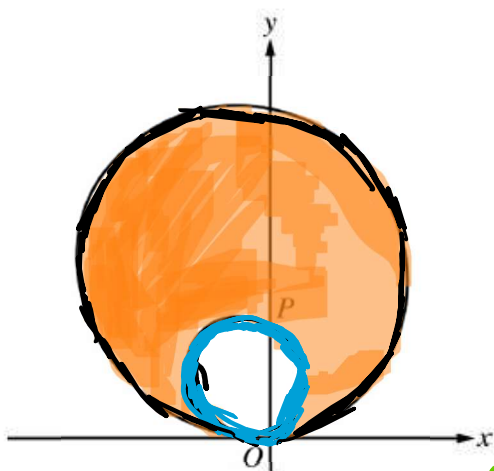
Symmetry ends when $3 \cos \theta = 0$, at $\theta = \pi/2$

The region switches the outside function when $3 \cos \theta = 2 - \cos \theta$
 $\cos \theta = 1/2, \theta = \pi/3$



$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/3} (2 - \cos \theta)^2 d\theta + 2 \cdot \frac{1}{2} \int_{\pi/3}^{\pi/2} (3 \cos \theta)^2 d\theta$$

Ex. 9: A graphing calculator is required for problems on this part of the exam.



A) $\frac{dx}{d\theta} = ? ?$ given r , we need $x \dots$

$$x = r \cos \theta = 3 \theta \sin \theta \cdot \cos \theta$$

$$\left. \frac{dx}{d\theta} \right|_{\theta = \pi/2} = \boxed{-4.712}$$

at $P, \theta = \pi/2$ * use nDeriv!!

Let r be the function given by $r(\theta) = 3\theta \sin \theta$ for $0 \leq \theta \leq 2\pi$. The graph of r in polar coordinates consists of two loops, as shown in the figure above. Point P is on the graph of r and the y -axis.

- (A) Find the rate of change of the x -coordinate with respect to θ at the point P .
 (B) Find the area of the region between the inner and outer loops of the graph.

B) No symmetry, so need to find bounds, which is when $r = 3\theta \sin \theta = 0$, that's is $\theta = 0, \pi, 2\pi$

Inner Loop is from $[0, \pi]$, Outer Loop is from $[\pi, 2\pi]$

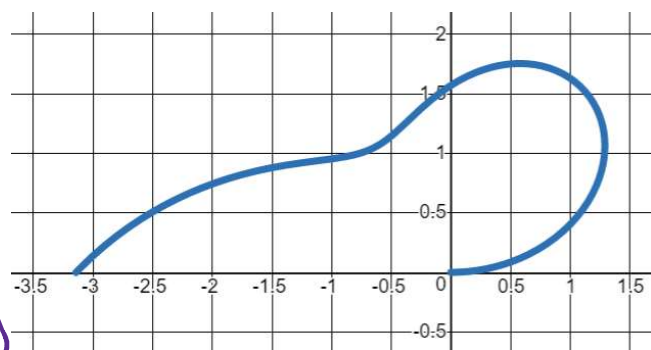
$$A = \frac{1}{2} \int_{\pi}^{2\pi} (r(\theta))^2 d\theta - \frac{1}{2} \int_0^{\pi} (r(\theta))^2 d\theta = \boxed{139.529}$$

Ex. 10: CA The curve shown below is drawn in the xy -plane and is described by the equation in polar coordinates $r = \theta + \sin 2\theta$ for $0 \leq \theta \leq \pi$, where r is measured in meters and θ is in radians.

- a) Find the area bounded by the curve and the x -axis.

$$A = \frac{1}{2} \int_0^{\pi} (\theta + \sin 2\theta)^2 d\theta$$

$$= \boxed{4.382}$$



- b) Find the angle θ that corresponds to the point on the curve with x -coordinate -2 .

$$x = r \cos \theta = (\theta + \sin 2\theta) \cos \theta = -2$$

$$\boxed{\theta = 2.786}$$

- c) For $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$, $\frac{dr}{d\theta}$ is negative. What does this fact say about r ? What does this fact say about the curve?

$\frac{dr}{d\theta} < 0$ means the radius is decreasing

Since r is decreasing, the curve is getting closer to the pole

- d) Find the value of θ in the interval $0 \leq \theta \leq \frac{\pi}{2}$ that corresponds to the point on the curve in the first quadrant with greatest distance from the origin. Justify your answer.

↓ r -value ... Absolute max of r

Candidates: endpoints $\Rightarrow \theta = 0, \frac{\pi}{2}$

critical values $\Rightarrow r' = 1 + 2\cos 2\theta = 0$

$$\theta = 1.04719755$$

$$r(0) = 0$$

$$\boxed{r(1.04719755) = 1.913}$$

$$r(\frac{\pi}{2}) = 1.571$$

Arc Length

When we want to find the arc length of a region in polar, we think of this parametrically. The only difference is the instead of t and dt , we will be using θ and $d\theta$. This gives you:

Rectangular: $\int_a^b \sqrt{\left(\frac{dx}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + (f'(x))^2} dx$

Parametric: $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

Polar: $\int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$ $x = r \cos \theta$
 $y = r \sin \theta$

Ex. 11: Find the length of the spiral $r = e^\theta$ between $\theta = 0$ and $\theta = \frac{\pi}{2}$.

$\int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$ $x = e^\theta \cos \theta$, $y = e^\theta \sin \theta$

$$\frac{dx}{d\theta} = e^\theta \cos \theta - e^\theta \sin \theta$$

$$\frac{dy}{d\theta} = e^\theta \sin \theta + e^\theta \cos \theta$$

$$= \int_0^{\pi/2} \sqrt{(e^\theta \cos \theta - e^\theta \sin \theta)^2 + (e^\theta \sin \theta + e^\theta \cos \theta)^2} d\theta$$

$$= \int_0^{\pi/2} \sqrt{e^{2\theta} \cos^2 \theta - 2e^{2\theta} \cos \theta \sin \theta + e^{2\theta} \sin^2 \theta + e^{2\theta} \sin^2 \theta + 2e^{2\theta} \sin \theta \cos \theta + e^{2\theta} \cos^2 \theta} d\theta$$

$$= \int_0^{\pi/2} \sqrt{2e^{2\theta} (\cos^2 \theta + \sin^2 \theta)} d\theta = \int_0^{\pi/2} \sqrt{2} e^\theta d\theta$$

$$= \sqrt{2} e^\theta \Big|_0^{\pi/2} = \boxed{\sqrt{2} (e^{\pi/2} - 1)}$$

Ex. 12: Find the length of the graph of $r = 3 \cos \theta$.

$$x = 3 \cos^2 \theta, \quad \frac{dx}{d\theta} = -6 \cos \theta \sin \theta$$

$$y = 3 \cos \theta \sin \theta$$

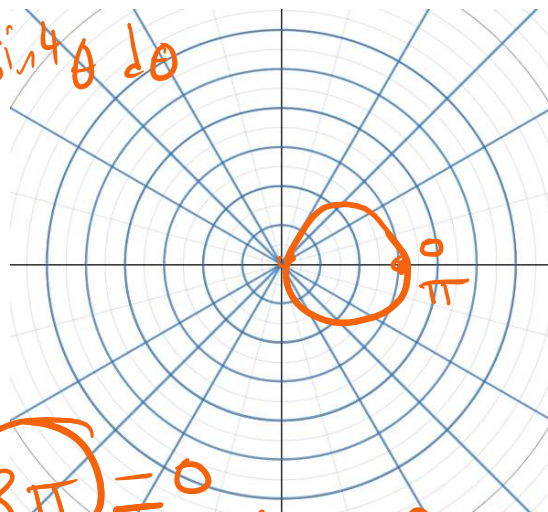
$$\frac{dy}{d\theta} = 3 \cos^2 \theta - 3 \sin^2 \theta$$

$$\int_0^\pi \sqrt{(-6 \cos \theta \sin \theta)^2 + (3 \cos^2 \theta - 3 \sin^2 \theta)^2} d\theta$$

$$\int_0^\pi \sqrt{36 \cos^2 \theta \sin^2 \theta + 9 \cos^4 \theta - 18 \cos^2 \theta \sin^2 \theta + 9 \sin^4 \theta} d\theta$$

$$\int_0^\pi \sqrt{9 (\cos^4 \theta + 2 \cos^2 \theta \sin^2 \theta + \sin^4 \theta)} d\theta$$

$$\int_0^\pi 3 \sqrt{(\cos^2 \theta + \sin^2 \theta)^2} d\theta = \int_0^\pi 3 d\theta = 3\theta \Big|_0^\pi = 3\pi$$



Ex. 13: Find the length of the polar curve $r = 2 \sec \theta$ from $\theta = 0$ to $\theta = \frac{\pi}{3}$.

circumference of circle

$$x = 2 \sec \theta \cos \theta = 2$$

$$\frac{dx}{d\theta} = 0$$

$$y = 2 \sec \theta \sin \theta = 2 \tan \theta$$

$$\frac{dy}{d\theta} = 2 \sec^2 \theta$$

$$\int_0^{\pi/3} \sqrt{(0)^2 + (2 \sec^2 \theta)^2} d\theta = \int_0^{\pi/3} |2 \sec^2 \theta| d\theta$$

since $\sec^2 \theta > 0$ from $0 < \theta < \pi/3$, we don't need abs.-value!

$$= 2 \int_0^{\pi/3} \sec^2 \theta d\theta = 2 \tan \theta \Big|_0^{\pi/3} = 2 \tan \pi/3 - 2 \tan 0 = \boxed{2\sqrt{3}}$$